# Project Report winter 2026

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***Abstract*—This report introduces a machine learning pipeline for classifying short news snippets into thematic categories, addressing challenges such as high data sparsity and class imbalance. The proposed approach consists of a hybrid feature extraction strategy that integrates TF-IDF textual vectors with structured metadata, specifically leveraging publisher information as a domain prior. A Linear Support Vector Classifier (LinearSVC) was optimized via grid search, achieving a Macro F1-score of 0.7148 on the validation set. The results demonstrate that fusing metadata with semantic features significantly outperforms naive baselines in short-text classification tasks.**

## I. Problem Overview

The objective of this project is to classify news articles collected from various international sources into seven predefined categories. The provided dataset (*development.csv*) contains unstructured textual fields (`title`, `article`) and structured metadata (`source`, `timestamp`, `page_rank`).

An exploratory data analysis revealed several critical challenges that informed the pipeline design:

1) **Class imbalance**: as shown in Figure 1, the dataset is heavily skewed. The majority class (*International*) constitutes approximately 29.4% of the data, while the minority class (*Health*) represents only 3.9%. This imbalance necessitates the use of the Macro F1-score as the primary evaluation metric, as it assigns equal weight to each category regardless of its frequency, preventing the model from biasing towards majority classes.

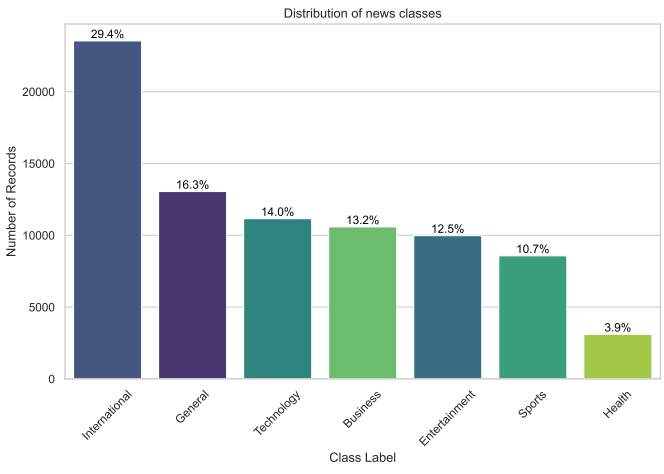



Fig. 1. Distribution of news categories.

2) **Short text length**: the textual content consists primarily of brief snippets rather than full articles. The average length is approximately 35 words, with a subset of outliers containing fewer than 5 words (Figure 2). These extremely short instances often lack sufficient semantic context for reliable classification based solely on text, requiring the integration of metadata.

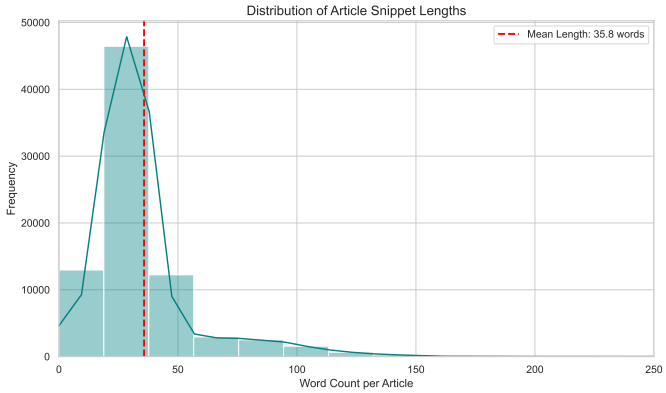



Fig. 2. Distribution of word counts per snippet. The red dashed line indicates the mean length (≈ 35 words).

3) **Label noise and inconsistency**: a duplicate check revealed over 5000 duplicated snippets. More critically, 1614 cases involve identical text associated with contradictory labels. This introduces an irreducible error into the dataset. Specifically, these ambiguities violate the assumption of a consistent mapping between inputs and targets, potentially preventing any classifier from achieving perfect accuracy on the training set.
4) **Data quality and missing values**: a quality check revealed that, while the page_rank feature is fully populated, the textual content contains a negligible amount of null values, and the timestamp attribute contains a large proportion of placeholders (approximately 35% are '0000-00-00 00:00:00').
5) **Source distribution bias**: the dataset aggregates news from 1359 unique providers, following a power law distribution where the top 3 sources account for nearly 45% of entries (Figure 3). A correlation analysis revealed that specific sources are highly predictive of certain labels, suggesting `source` is a highly discriminative feature.

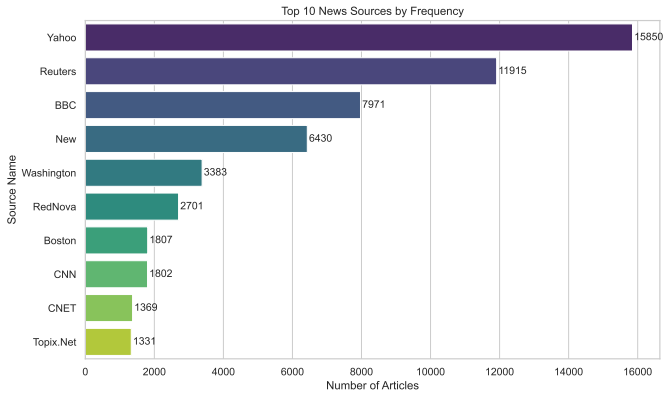



Fig. 3. Top 10 news sources. The distribution is heavily skewed towards Yahoo, Reuters, and BBC.

## II. Proposed approach

### A. Preprocessing

To capture the full semantic context, textual data (`source`, `title`, and `article`) were merged into a single unified field. Operating under the hypothesis that the publisher acts as a strong domain prior and the title contains the densest information, the `source` and `title` tokens were up-weighted relative to the body text. Specifically, these components were repeated three times prior to concatenation. This heuristic artificially boosts their term frequency, ensuring that these highly discriminative signals carry significantly greater weight during the subsequent vectorization phase. The merged text was then normalized via HTML decoding, removal of non-alphabetic characters and lowercasing. We applied a *Snowball Stemmer* to collapse inflectional forms to their root, significantly reducing the dimensionality of the vocabulary. The cleaned text was vectorized using *TF-IDF* [1] with an $n$-gram range of $[1, 2]$ to capture short phrases. To manage the high dimensionality, a $\chi^2$ (Chi-Squared) statistical test was applied via `SelectKBest` to retain only the top $k$ most informative features.

We addressed data quality issues through strict filtering criteria:

- **Length filtering**: snippets shorter than 20 characters were discarded, as they lacked sufficient semantic content for discriminative power.
- **Duplicate management**: the EDA revealed significant redundancy. We adopted a rigorous resolution strategy to sanitize the training set:
  1) *Exact duplicates*: redundant instances (identical title, article and label) were consolidated to a single occurrence to prevent overfitting and reduce computational cost.
  2) *Conflicting duplicates*: for identical snippets (equal title and article) with different labels, we applied a majority voting mechanism. We assigned the text to the dominant class to enforce consistency, while retaining the option to discard ambiguous instances entirely in cases of irresolvable ties (no majority), thus preventing the introduction of incoherent training signals.

Missing textual fields were imputed with empty strings to maintain data integrity. The `timestamp` attribute, in contrast, required a more specialized approach, as approximately 35% of entries contained invalid placeholders ('0000-00-00'). To leverage this information rather than treating it solely as noise, we engineered three specific features:

- **has_date**: a binary flag indicating the presence of a valid timestamp. This captures potential structural differences in data quality across sources.
- **day_of_week**: extracted from valid timestamps to capture cyclical publication patterns (e.g., business news on weekdays, sports on weekends). Invalid entries were imputed with a sentinel value (-1) prior to encoding. This ensures that the absence of a date is treated as a distinct, informative category rather than noise.
- **hour**: represents the publication hour (0-23). Missing values were explicitly imputed with -1 rather than 0. This distinction is critical to ensure the model does not conflate unknown times with midnight publications (00:00).

Finally, distinct transformations were applied to metadata to ensure compatibility with the sparse text vectors:

- **Numerical features**: we applied a `MinMaxScaler` to `page_rank`, `hour` and `has_date` features. This transformation maps values to the $[0, 1]$ interval. It is critical for two reasons: first, it prevents features with larger magnitudes (such as `hour`) from dominating the SVM optimization objective; second, for `page_rank`, it handles the strictly non-Gaussian distribution by preserving the relative distances without enforcing a normal distribution (as StandardScaler would). s illustrated in Figure 4
- **Categorical features**: the source distribution exhibits a clear power-law behavior (Figure 5). We performed a `OneHotEncoder` with the `min_frequency` parameter to aggregate rare tail-end sources into a single *infrequent* category, maintaining a compact feature space and reducing memory consumption.

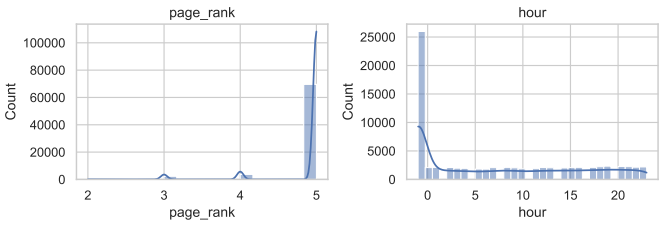



Fig. 4. Distribution of Numerical Features.

### B. Model selection

To handle the high-dimensional sparse vectors and the intrinsic label noise identified during the exploratory analysis, we evaluated two discriminative models:

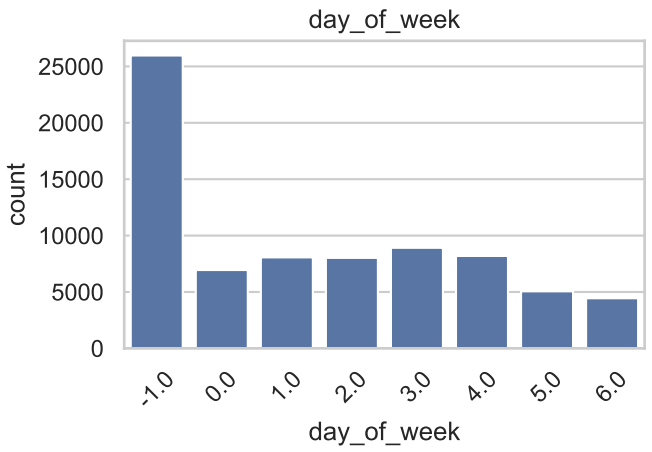



Fig. 5. Distribution of `day_of_week` feature

- **Logistic Regression**: selected as a baseline due to its interpretability and efficiency on sparse textual data. However, preliminary experiments indicated a performance ceiling (Cross-Validation Macro F1 $\approx 0.69$), suggesting the model struggled to separate semantically overlapping categories (as *General News* and *International News*) given the limited context of the short text snippets.
- **Linear Support Vector Classifier (LinearSVC)**: selected as the final model, SVMs [2] are inherently robust in high-dimensional spaces where the number of features is significant relative to the number of samples. This model consistently outperformed the baseline by finding a hyperplane that maximizes the margin between classes, offering a more stable separation for the heterogeneous feature set (TF-IDF vectors combined with dense metadata).

We specifically configured the LinearSVC to address the unique challenges of the provided dataset.

1) **Robustness to label noise**: as mentioned above, the dataset contains over 1600 instances with contradictory labels. To address this, we leveraged the **soft-margin** formulation of the LinearSVC. The regularization parameter $C$ governs the trade-off between maximizing the margin width and minimizing classification errors. By selecting a low value ($C = 0.2$), we reduced the penalty assigned to margin violations. This configuration allows the model to tolerate misclassifications on noisy outliers rather than overfitting the decision boundary to conflicting data points, thus prioritizing better generalization.
2) **Optimization for Macro F1-Score**: to mitigate the bias induced by the severe class imbalance, we adopted a cost-sensitive learning strategy. By setting class weights inversely proportional to their frequencies, the SVM effectively assigns a higher penalty to misclassifications of minority instances. This prevents the optimization algorithm from converging to trivial solutions that maximize global accuracy, but fail to capture underrepresented categories. Thereby, the Macro F1 metric is directly improved.
3) **Handling feature heterogeneity**: the model proved exceptional stability when fusing in a high-dimensional joint feature space. It successfully integrated 40000 sparse TF-IDF vectors with dense structured metadata. Since the publisher identity acts as a strong domain prior, the SVM's ability to linearly combine these heterogeneous signals was decisive in achieving the final Validation Macro F1-score.

### C. Hyperparameters Tuning

The optimization process addressed the entire pipeline, not only the classifier. We performed a `GridSearchCV` with 3-fold cross-validation.

The search space included the following key components:

- **TfidfVectorizer (n-gram range)**: we assessed the impact of expanding the context window by comparing unigrams against bigrams ($n$-gram range: (1,1) vs (1,2)). Including bigrams proved crucial for disambiguating terminology in the *Technology* and *Business* categories. By capturing compound expressions (as "cloud computing" instead of "cloud"), the model reduced false positives caused by polysemous unigrams, thereby improving precision.
- **TfidfVectorizer (min_df)**: we examined minimum document frequencies (`min_df`) of [3, 5] to filter out terms appearing in negligible document subsets.Setting this threshold proved effective in reducing noise from rare words or typos, ensuring the vocabulary remained representative of the overall corpus.
- **SelectKBest**: we tuned the number of features $k$ within the range [30,000, 50,000]. Selecting $k = 40,000$ features via the $\chi^2$ test offered the best trade-off, filtering out terms with negligible discriminative power while preserving the vocabulary diversity needed for the minority classes.
- **LinearSVC Regularization ($C$)**: we tested regularization strengths in the interval $[0.1, 0.3]$. A value of $C = 0.2$ was selected as lower values of $C$ enforce a stronger regularization (wider margin). This soft-margin behavior allows the model to tolerate misclassifications on the noisy or contradictory training samples identified during preprocessing, effectively preventing overfitting to outliers.

The tuning of the parameters is summarized in Table I.

TABLE I
HYPERPARAMETER CONSIDERED

| Component | Parameter | Explored Values |
|---|---|---|
| TfidfVectorizer | ngram_range | [(1, 1), (1, 2)] |
| TfidfVectorizer | min_df | [3, 5] |
| SelectKBest | k | [30000, 40000, 50000] |
| LinearSVC | C | [0.1, 0.2, 0.3] |

## III. Results

The performance of the final pipeline was evaluated using a held-out validation set (20% of the development data) to ensure an unbiased estimate of the model's generalization capability. The final configuration achieved a **Validation Macro F1-score of 0.7148**. To better understand the model behavior, we analyzed the results in terms of *Precision* and *Recall*.

A detailed look at the classification report reveals significant variance in performance across different news categories:

- **High-F1 Categories (Sports and Technology)**: the model showed high proficiency in identifying *Sports* ($F1 : 0.86$) and *Technology* ($F1 : 0.82$). These classes benefit from a very high *Precision*, as they rely on a highly specific vocabulary (e.g., team names, technical jargon) and strong associations with niche publishers like CNET.
- **Semantic Overlap:** The lowest performance was observed in *General News* ($F_1 = 0.52$) and *Entertainment* ($F_1 = 0.59$). These categories suffer from low precision due to significant semantic overlap with other classes. For instance, generic terms in "General News" are frequently misclassified as "International" or "Business", as the short text snippets often lack sufficient context for disambiguation.
- **Minority Class Performance (Health)**: despite being the least represented category ($\approx 4\%$ of the data), *Health* achieved a respectable $F1 : 0.67$. This is a clear result of the `class_weight='balanced'` strategy: the model achieved a very high *Recall* (0.84), meaning it successfully captured most health-related articles, although at the cost of some *Precision* (false positives from other categories).

TABLE II
DETAILED CLASSIFICATION REPORT ON THE VALIDATION SET.

| Class | Precision | Recall | F1-score | Support |
|---|---|---|---|---|
| 0 (International) | 0.77 | 0.75 | 0.76 | 4509 |
| 1 (Business) | 0.74 | 0.82 | 0.78 | 2042 |
| 2 (Technology) | 0.81 | 0.84 | 0.82 | 2155 |
| 3 (Entertainment) | 0.65 | 0.53 | 0.59 | 1818 |
| 4 (Sports) | 0.79 | 0.94 | 0.86 | 1677 |
| 5 (General News) | 0.60 | 0.47 | 0.52 | 2352 |
| 6 (Health) | 0.56 | 0.85 | 0.67 | 574 |

To further investigate the classification errors, we analyzed the confusion matrix (shown in Figure 6). It highlights three key patterns:

1) **International News**: due to its lexical breadth, this category captures ambiguous instances, resulting in 825 *General News* and 454 *Entertainment* false positives.
2) **Semantic ambiguity**: a bidirectional overlap exists between *Business* and *Technology* (245 combined errors), reflecting the shared lexicon of corporate tech news.
3) **Minority class success**: the distinct diagonal for *Health* validates the class-weighting strategy, which ensured high recall despite the negligible support.

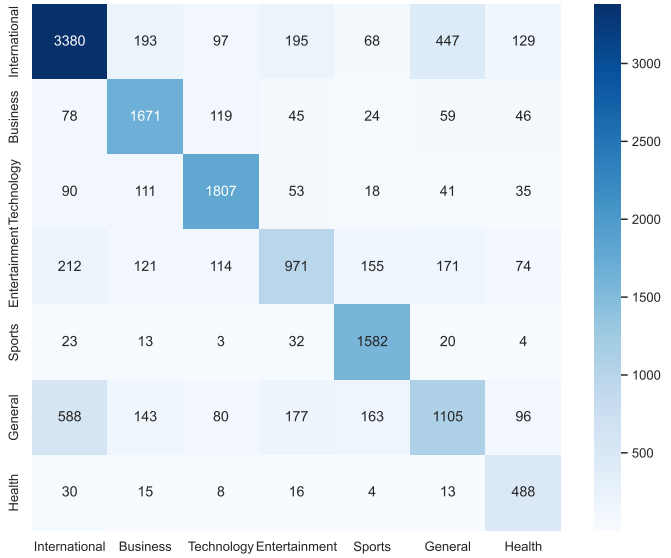



Fig. 6. Confusion matrix of the LinearSVC model on the validation set. The rows represent the true labels, while the columns represent the predicted labels. Note the significant off-diagonal density in Column 0 (International News).

## IV. Discussion

The proposed approach obtains results that outperform the naive baseline defined. It does so by leveraging a comprehensive preprocessing pipeline and a multi-modal feature engineering strategy that combines textual data with structural and temporal metadata. A pivotal factor in this success was the specific weighting strategy applied during preprocessing to the *title* and *source* tokens, which effectively forced the classifier to prioritize the high-density information fields over the noisier body text. More specifically, the experimental results demonstrate that while the textual content remains the primary driver for classification, the integration of structured metadata is essential for achieving high Macro F1-scores in this specific domain.

However, the following are some aspects that might be worth considering to further improve the obtained results:

- Source Reliability Metadata: integrating external information about the source could provide additional context that the current One-Hot encoding cannot capture.
- Strategies to counter the dataset's strong imbalance should be explored. These techniques, such as SMOTE [3], would help the model generalize better on minority classes, ultimately increasing the overall macro-F1 score.
- Replacing the TF-IDF representation with pre-trained word embeddings or transformer-based models would allow the system to understand the semantic context and synonyms, which are currently missed by the bag-of-words approach.

## References

[1] G. Salton and C. Buckley, "Term-weighting approaches in automatic text retrieval," *Information Processing & Management*, vol. 24, no. 5, pp. 513–523, 1988.

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[3] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "Smote: synthetic minority over-sampling technique," *Journal of artificial intelligence research*, vol. 16, pp. 321–357, 2002.